Multi-Agent Reinforcement Learning for Stag Hunt Games via Heterogeneous-Agent Proximal Policy Optimization and Cooperative Reward Reformulation

Giuseppe Maffucci
giuseppe.maffucci@example.com

Abstract

Cooperative multi-agent environments with simultaneous decision-making, such as the classic Stag Hunt game, present significant challenges due to non-stationarity and strong agent payoff dependencies. Standard Multi-Agent Reinforcement Learning (MARL) algorithms often fail when treating co-existing agents as part of a stochastic environment. To address this non-stationarity and mutual reward influence, we apply Heterogeneous-Agent Proximal Policy Optimization (HAPPO), leveraging its Advantage Decomposition Lemma to perform sequential agent updates with monotonicity guarantees on joint policy improvement. Furthermore, since HAPPO relies on a shared cooperative objective, we reformulate individual agent rewards into a unified global cooperative reward across three distinct game environments: Hunt, Harvest, and Escalation. In the Escalation scenario, we incorporate a Manhattan distance-based shaping term to overcome sparse rewards and improve target reachability during early training. Empirical evaluations demonstrate that our approach successfully fosters coordination across all scenarios, achieving rapid convergence.

1 Introduction

1.1 Context & Motivation

Cooperative decision-making in multi-agent systems is fundamentally constrained by the tension between individual risk and collective benefit. In classical game theory, matrix games such as the Stag Hunt model this tension through a stark trade-off: agents must choose between a payoff-dominant equilibrium (cooperatively hunting a stag) that yields maximum reward only through mutual trust and simultaneous coordination, and a risk-dominant equilibrium (hunting a hare) that guarantees a safer, suboptimal payoff (Skyrms, 2004). Transitioning from this default "state of nature" to a social contract requires full mutual belief, as miscoordination or desertion by a single participant leads to zero payoff for cooperative agents. In spatial and grid-world environments, this dynamic expands into Sequential Social Dilemmas (SSDs), where decisions are no longer atomic or instantaneous, but instead unfolded over time under partial observability (Leibo et al., 2017). In these temporally extended settings, such as wolfpack hunting scenarios, achieving optimal joint payoffs places severe demands on policy coordination, where non-stationarity and the cognitive complexity of executing joint trajectories often push independent learning agents toward risk-dominant, non-cooperative baselines.

1.2 Problem Statement & Non-Stationarity

Policy Gradient (PG) methods and trust region formulations such as PPO (Schulman et al., 2017) offer theoretical guarantees of monotonic policy improvement in single-agent RL. However, naive application of independent PG algorithms to multi-agent environments frequently fails . When multiple agents update their policies simultaneously, treating co-existing agents merely as stochastic components of a stationary environment breaks underlying convergence assumptions. Because all agents alter their behavior concurrently, the optimization landscape shifts dynamically for each individual learner, resulting in severe non-stationarity, high variance, and training instability (Kuba et al., 2022). Although Centralized Training with Decentralized Execution (CTDE) frameworks—such as MAPPO (**?**)—mitigate this by conditioning value functions on joint observations, simultaneous update schemes still lack theoretical guarantees for joint monotonic improvement (Kuba et al., 2022). In coordination-critical games like the Stag Hunt, simultaneous updates often push agents toward suboptimal, risk-dominant equilibria.

1.3 Proposed Solution

To overcome this fundamental limitation, Heterogeneous-Agent Proximal Policy Optimization (HAPPO) introduces a sequential update scheme backed by the Multi-Agent Advantage Decomposition Lemma (Kuba et al., 2022). By updating agent policies sequentially, HAPPO explicitly accounts for inter-agent dependencies during step-by-step advantage updates, theoretically guaranteeing joint policy improvement ($J(\boldsymbol{\pi}') \geq J(\boldsymbol{\pi})$) and preventing miscoordination collapse.

To align with HAPPO's unified optimization objective, we reformulate individual agent payoffs into a joint, cooperative reward scheme evaluated across three distinct game variants: Hunt, Harvest, and Escalation. Furthermore, in the Escalation scenario, we incorporate a spatial distance-based potential term to mitigate reward sparsity and guide early policy exploration.

2 Background & Theoretical Framework

2.1 Cooperative Markov Games

Following Kuba et al. (2022), we model the multi-agent decision process as a fully cooperative Markov Game defined by the tuple $\langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$, where $\mathcal{N} = \{1, \dots, n\}$ denotes the set of agents, $\mathcal{S}$ is the finite state space, and $\mathcal{A} = \prod_{i=1}^{n} \mathcal{A}^i$ represents the joint action space (where $\mathcal{A}^i$ is the local action space for agent i). The transition probability function is $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$, $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the shared reward function, and $\gamma \in [0, 1)$ is the discount factor. At time step $t \in \mathbb{N}$, given state $s_t \in \mathcal{S}$, each agent i selects an action $\mathrm{a}_t^i \in \mathcal{A}^i$ according to its local policy $\pi^i(\cdot|s_t)$, forming a joint action $\mathbf{a}_t = (\mathrm{a}_t^1, \dots, \mathrm{a}_t^n) \sim \boldsymbol{\pi}(\cdot|s_t) = \prod_{i=1}^{n} \pi^i(\cdot|s_t)$. The environment transitions to s_{t+1} according to probability $P(s_{t+1}|s_t, \mathbf{a}_t)$ and yields a shared cooperative reward $r_t = r(s_t, \mathbf{a}_t) \in \mathbb{R}$. We define the (improper) marginal state distribution as $\rho_{\boldsymbol{\pi}} \triangleq \sum_{t=0}^{\infty} \gamma^t \rho_{\pi}^t$. All agents share the same objective: to maximize the expected discounted total return, defined as:

$$J(\boldsymbol{\pi}) \triangleq \mathbb{E}_{s_{0:\infty} \sim \rho_{\boldsymbol{\pi}}^{0:\infty},\, \mathbf{a}_{0:\infty} \sim \boldsymbol{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \tag{1}$$

2.2 Advantage Decomposition Lemma

Standard multi-agent policy gradient methods suffer from severe non-stationarity when updating agents simultaneously. To address this, HAPPO leverages the Multi-Agent Advantage Decomposition Lemma (Kuba et al., 2022), which proves that the joint advantage function over an ordered subset of agents $i_{1:m}$ can be evaluated as a sum of local sequential advantages:

$$A_{\boldsymbol{\pi}}^{i_{1:m}} \left(s, \boldsymbol{a}^{i_{1:m}} \right) = \sum_{j=1}^{m} A_{\boldsymbol{\pi}}^{i_j} \left(s, \boldsymbol{a}^{i_{1:j-1}}, a^{i_j} \right) \tag{2}$$

This lemma guarantees that if each agent sequentially updates its policy such that its local advantage remains positive, the overall joint policy performance $J(\boldsymbol{\pi})$ improves monotonically.

2.3 HAPPO Policy Update Scheme

HAPPO operationalizes this sequential decomposition by randomly sampling a permutation of agents $i_{1:n}$ at each iteration. For a specific agent i_m, the local objective is optimized using a modified PPO-Clip formulation:

$$L(\theta^{i_m}) = \hat{\mathbb{E}}_t \left[\min \left(\frac{\pi^{i_m}_{\theta^{i_m}}(a^{i_m}_t | o^{i_m}_t)}{\pi^{i_m}_{\theta^{i_m}_k}(a^{i_m}_t | o^{i_m}_t)} M^{i_{1:m}}(s_t, \boldsymbol{a}_t), \text{ clip} \left(\frac{\pi^{i_m}_{\theta^{i_m}}(a^{i_m}_t | o^{i_m}_t)}{\pi^{i_m}_{\theta^{i_m}_k}(a^{i_m}_t | o^{i_m}_t)}, 1 \pm \epsilon \right) M^{i_{1:m}}(s_t, \boldsymbol{a}_t) \right) \right] \tag{3}$$

where $M^{i_1}(s_t, \boldsymbol{a}_t) = \hat{A}(s_t, \boldsymbol{a}_t)$ for the first agent in the sequence, and the factor M is updated sequentially for subsequent agents as:

$$M^{i_{1:m+1}}(s_t, \boldsymbol{a}_t) = \frac{\pi^{i_m}_{\theta^{i_m}_{k+1}}(a^{i_m}_t | o^{i_m}_t)}{\pi^{i_m}_{\theta^{i_m}_k}(a^{i_m}_t | o^{i_m}_t)} M^{i_{1:m}}(s_t, \boldsymbol{a}_t) \tag{4}$$

This sequential integration ensures that agent i_{m+1} evaluates its policy shift conditioned on the updated decisions of all preceding agents $i_{1:m}$.

3 Methodology & Environment Scenarios

To facilitate joint coordination in simultaneous decision-making tasks, we adapt the Stag Hunt paradigm into three distinct game scenarios: *Hunt*, *Harvest*, and *Escalation*.

3.1 Cooperative Reward Reformulation

HAPPO's theoretical guarantees on joint monotonic policy improvement rely on a shared optimization objective across all participating agents. Consequently, individual agent payoffs R_i cannot be optimized independently without breaking the joint advantage bounds. To align the local environment objectives with HAPPO's requirements, we reformulate the task into a fully cooperative setting by defining a global cooperative reward function R_{coop} aggregated over all n agents:

$$R^{i_{1:n}}_{\text{coop},t} = \sum_{m=1}^{n} R^{i_m}_t \tag{5}$$

3.2 Environment Scenarios & Reward Design

Hunt Mode. In the *Hunt* scenario, agents must simultaneously execute a capture action on the target to receive a positive reward. The global cooperative reward at timestep t is calculated directly as the sum of individual agent interactions:

$$R^{i_{1:n}(\text{Hunt})}_{\text{coop},t} = \sum_{m=1}^{n} R^{i_m}_t \tag{6}$$

Harvest Mode. The *Harvest* scenario involves resource collection under spatial competition and coordination constraints. Similar to the *Hunt* setting, the global objective is maintained via additive reward pooling:

$$R^{i_{1:n}(\text{Harvest})}_{\text{coop},t} = \sum_{m=1}^{n} R^{i_m}_t \tag{7}$$

Escalation Mode & Potential-Based Reward Shaping. The *Escalation* scenario introduces severe temporal and spatial coordination challenges. Agents must first navigate to a sleeping target (*Mark*) simultaneously. The target remains stationary until both agents reach its grid position, after which it attempts to escape, requiring agents to maintain a synchronized trajectory (occupying adjacent or identical cells) for K consecutive timesteps.

Due to the initial phase requiring simultaneous arrival at the target, unshaped environmental rewards suffer from extreme sparsity ($R^{i_m,\text{env}}_t$), leading to inefficient random exploration. To address this, we introduce a Potential-Based Reward Shaping (PBRS) mechanism. Let $p_t^{i_m} = (x_t^{i_m}, y_t^{i_m})$ denote the grid coordinates of agent $i_m \in \{i_1, \dots, i_n\} = i_{1:n}$ and $p_t^M = (x_t^M, y_t^M)$ denote the position of the target at time t. We define the spatial state potential $\Phi(s_t)$ as the total Manhattan distance of all agents to the target:

$$\Phi(s_t) = \sum_{i=m}^{n} \left(|x_t^{i_m} - x_t^M| + |y_t^{i_m} - y_t^M| \right) \tag{8}$$

The shaped potential difference F_t between consecutive states s_t and s_{t+1} is formulated as:

$$F_t = \Phi(s_t) - \gamma_{\text{s}} \Phi(s_{t+1}) \tag{9}$$

where $\gamma_{\text{s}} \in (0, 1)$ is the reward shaping factor. The final augmented global reward $R_{\text{coop},t}^{\text{Escalation}}$ combines the base environmental payoff $R_{\text{coop},t}^{i_{1:n}\text{env}} = \sum_{m=1}^{n} R_t^{i_m,\text{env}}$ with the weighted potential gradient, leading to:

$$R_{\text{coop},t}^{i_{1:n}(\text{Escalation})} = \sum_{m=1}^{n} R_t^{i_m,\text{env}} + \alpha_{\text{s}} F_t(s_t, s_{t+1}) = \sum_{m=1}^{n} R_t^{i_m,\text{env}} + \alpha_{\text{s}} \left(\Phi(s_t) - \gamma_{\text{s}} \Phi(s_{t+1}) \right) \tag{10}$$

where $\alpha > 0$ is a scaling hyperparameter. By grounding F_t in potential differences, this formulation preserves the policy invariance properties of optimal policy trajectories while significantly accelerating convergence during early training.

3.3 Entropy

In the objective implementation, in addition to $L(\theta^{i_m})$ we consider also an entropy term, leading to a total objective: $L_{\text{HAPPO}}(\theta^{i_m}) = L(\theta^{i_m}) + \beta_{\text{H}} * H(\theta^{i_m})$ where H is the entropy for that specific agent at that specific iteration k

4 Experimental Setup & Results

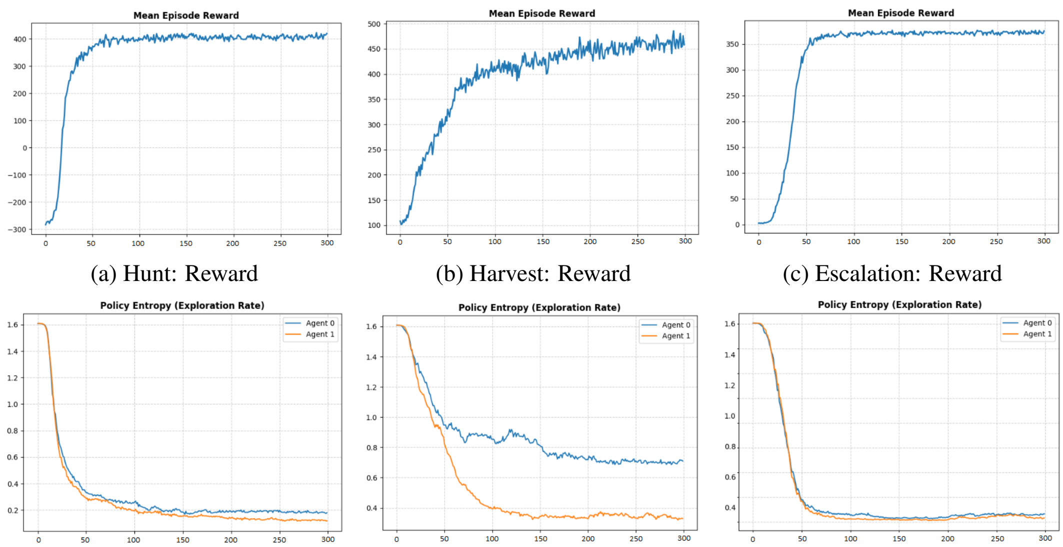
4.1 Experimental Setup & Hyperparameters

Both actor and critic networks are built upon Multi-Layer Perceptron (MLP) architectures. Specifically, the networks consist of three hidden linear layers with ReLU activation functions. The critic outputs a scalar value representing the joint state value, while each actor network outputs a 5-dimensional categorical distribution corresponding to the discrete action space of the environment.

The primary training hyperparameters utilized across all experiments are summarized in Table 1.

Table 1: Hyperparameter configurations for HAPPO training (same for Hunt, Harvest, Escalation).

Hyperparameter	**Value**
Optimization Iterations (K)	200
Warmup Iterations (K_{warmup})	10 (5% of K)
Actor Learning Rate (LR_{actor})	3×10^{-4}
Critic Learning Rate ($\text{LR}_{\text{critic}}$)	1×10^{-3}
Actor Epochs	3
Critic Epochs	5
Num. Mini-batches	4
PPO Clip Parameter (ϵ)	0.2
Value Loss Coefficient	0.5
Entropy Coefficient (β_{H})	0.01
Max Gradient Norm	0.5
Use Clipped Value Loss	False
Huber Loss Delta	10.0
Discount Factor GAE (γ)	0.99
Discount Factor Reward Shaping (γ_{s})	0.99
Reward Shaping Weight (α_{s})	0.1



(a) Hunt: Reward (b) Harvest: Reward (c) Escalation: Reward

(d) Hunt: Entropy (e) Harvest: Entropy (f) Escalation: Entropy

Figure 1: Training dynamics across modes. Top row: Mean Episode Reward showing rapid convergence to optimal collective payoffs. Bottom row: Policy Entropy illustrating the transition from initial uniform exploration (≈ 1.61) to specialized deterministic policies.

4.2 Performance Analysis & Training Dynamics

As illustrated in the empirical learning curves in Figure 1, the MLP-based HAPPO implementation demonstrates steep early reward growth across all three modes (*Hunt*, *Harvest*, and *Escalation*). In the *Escalation* scenario, the inclusion of potential-based reward shaping successfully overcomes initial reward sparsity, enabling agents to rapidly learn target-reaching policies. Across all environments, performance stabilizes smoothly near optimal collective reward plateaus, validating the efficacy of sequential advantage updates in establishing robust multi-agent coordination.

5 Discussion & Limitations

While our results demonstrate that HAPPO effectively resolves simultaneous Stag Hunt dilemmas, certain limitations remain. Principally, the underlying grid-world environments feature relatively low-dimensional state spaces and fully observable grid dynamics.

6 Conclusion

In this work, we addressed the non-stationarity and miscoordination challenges inherent in simultaneous Stag Hunt games by applying Heterogeneous-Agent Proximal Policy Optimization (HAPPO). By reformulating individual agent payoffs into a centralized, cooperative reward structure and integrating potential-based spatial reward shaping for complex scenarios, our approach guarantees joint monotonic policy improvement. Empirical results confirm fast convergence and stable coordination across all tested environments laying a foundation for future integrations with autoregressive sequential models.

References

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